# The energy-score loss for a one-step transport map: derivation, a check of the proposed formula, and the minibatch estimator

Stage 1 of the `transport_map` pipeline (written by an unattended Claude agent)

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**Abstract**

We derive the training loss for a map $F_\theta$ that pushes Gaussian noise to the data in one forward pass, starting from the definition of the energy score in Gneiting and Raftery [1]. The loss is

$$\mathcal{L}(\theta) = \mathbb{E}\,\|F_\theta(Z) - Y\| - \tfrac{1}{2}\,\mathbb{E}\,\|F_\theta(Z) - F_\theta(Z')\|, \qquad Z, Z' \overset{\text{iid}}{\sim} \mathcal{N}(0, I),\ Y \sim Q,$$

with $Z$, $Z'$ and $Y$ mutually independent. It equals half the energy distance between model and data plus a constant, so its unique minimiser is the data distribution. The class-conditional version keeps this property if the second term pairs two samples *of the same class*. The formula proposed in the project's `CLAUDE.md` has the right first term, sign and factor $\frac{1}{2}$. Its second term is wrong as typed: $\mathbb{E}[F_\theta(x) - F_\theta(x')]$ is a raw difference whose expectation is 0, so the proposed loss reduces to $\mathbb{E}\|F_\theta(x) - y\|$. That loss is minimised by a point mass at the geometric median of the data, i.e. complete mode collapse. The fix is to take the norm, $\|F_\theta(x) - F_\theta(x')\|$, with $x, x'$ drawn independently from the *source Gaussian*. Proofs, primary-source citations and CPU checks (Section 9) support each step.

## Contents

**1 Setup** **2**

**2 Scoring rules** **2**

**3 The energy score** **2**

**4 Expected score minus its minimum is half the energy distance** **4**

**5 The loss for the pushforward model** **4**

**6 Line-by-line check of the proposed formula** **6**

**7 Minibatch estimator** **8**

**8 Related work** **10**

**9 Numerical checks** **11**

**10 Summary for the next stages** **17**

# 1 Setup

- **Data.** $Q$ is a Borel probability measure on $\mathbb{R}^d$; an image is flattened to a vector $y \in \mathbb{R}^d$ (e.g. $d = 3 \cdot 32 \cdot 32$). $Y, Y' \sim Q$ denote independent copies. With labels, $(Y, C) \sim Q$ on $\mathbb{R}^d \times \{1, \dots, K\}$, with class probabilities $\pi_c = Q(C = c)$ and class-conditional laws $Q_c = \mathcal{L}(Y \mid C = c)$.
- **Source.** $\mu = \mathcal{N}(0, I_k)$ on $\mathbb{R}^k$. For DiT $k = d$ (noise of image shape), but nothing below needs $k = d$.
- **Map and model.** $F_\theta : \mathbb{R}^k \to \mathbb{R}^d$ (or $F_\theta : \mathbb{R}^k \times \{1..K\} \to \mathbb{R}^d$), measurable for every parameter $\theta$. The model is the pushforward $P_\theta = (F_\theta)_\# \mu$, i.e. $P_\theta(A) = \mu(F_\theta^{-1}(A))$; conditionally $P_\theta^c = (F_\theta(\cdot, c))_\# \mu$. For any measurable $g \geq 0$ or integrable $g$, $\mathbb{E}_{X \sim P_\theta} g(X) = \mathbb{E}_{Z \sim \mu} g(F_\theta(Z))$ (change of variables for image measures). So every expectation over the model can be computed by sampling $Z$ and evaluating $F_\theta$: no density of $P_\theta$, no invertibility of $F_\theta$ and no $k = d$ are needed.
- **Norm and moments.** $\|\cdot\|$ is the Euclidean norm. For $\beta > 0$, $\mathcal{P}_\beta$ is the set of Borel probability measures $P$ on $\mathbb{R}^d$ with $\mathbb{E}_P \|X\|^\beta < \infty$.

# 2 Scoring rules

Gneiting and Raftery [1, Sec. 1–2] take a scoring rule to be a reward $S^{\mathrm{GR}}(P, y)$ that the forecaster wishes to *maximise* ("positively oriented") when the forecast is $P$ and $y$ materialises. They write $S^{\mathrm{GR}}(P, Q) = \int S^{\mathrm{GR}}(P, y)\, dQ(y)$ for the expected score under $Q$, and call $S^{\mathrm{GR}}$ *proper* relative to a class $\mathcal{P}$ if $S^{\mathrm{GR}}(Q, Q) \geq S^{\mathrm{GR}}(P, Q)$ for all $P, Q \in \mathcal{P}$ [1, eq. (1)], and *strictly proper* if, in addition, equality holds only when $P = Q$.

**Our sign convention (fixed for the rest of this note).** We train by minimisation, so we use the *negatively oriented* score $S = -S^{\mathrm{GR}}$ throughout. With this convention:

**Definition 1.** A scoring rule $S : \mathcal{P} \times \mathbb{R}^d \to \mathbb{R}$ (smaller is better) with expected score $S(P, Q) = \mathbb{E}_{Y \sim Q} S(P, Y)$ is *proper* relative to $\mathcal{P}$ if $S(Q, Q) \leq S(P, Q)$ for all $P, Q \in \mathcal{P}$, and *strictly proper* if moreover $S(P, Q) = S(Q, Q)$ implies $P = Q$. The *divergence* is $d(P, Q) = S(P, Q) - S(Q, Q) \geq 0$ ([1, eq. (7)] in their orientation).

Why this matters for learning: if $Q \in \{P_\theta\}$ and $S$ is strictly proper, then $\theta \mapsto S(P_\theta, Q)$ is minimised exactly at those $\theta$ with $P_\theta = Q$. Only the expected score under $Q$ is needed, and that can be estimated from data samples. Adding to $S$ a function $h(y)$ of the observation alone, or multiplying by $c > 0$, preserves (strict) propriety [1, eq. (2)]; we use this below to drop constants.

# 3 The energy score

**Definition 2** (Energy score [1, Sec. 4.3, eq. (22)])**.** For $\beta \in (0, 2)$ and $P \in \mathcal{P}_\beta$, Gneiting and Raftery define

$$\mathrm{ES}_\beta(P, y) = \tfrac{1}{2} \mathbb{E}_P \|X - X'\|^\beta - \mathbb{E}_P \|X - y\|^\beta,$$

where $X, X'$ are independent with law $P$; it is positively oriented. Our negatively oriented version is

$$S_\beta(P, y) \;=\; -\mathrm{ES}_\beta(P, y) \;=\; \mathbb{E}_P \|X - y\|^\beta - \tfrac{1}{2} \mathbb{E}_P \|X - X'\|^\beta. \tag{1}$$

They state that the energy score is strictly proper relative to $\mathcal{P}_\beta$, by Theorem 1 of Székely (2003) (as cited in [1]), and that for the limiting case $\beta = 2$ it reduces to the negative squared error $\mathrm{ES}(P, x) = -\|\mu_P - x\|^2$, which is "regular and proper, but not strictly proper, relative to the class $\mathcal{P}_2$" [1, eq. (23)]. In dimension $d = 1$ with $\beta = 1$ it is the CRPS. We now prove these claims.

**Finiteness.** For $a, b \geq 0$, $(a+b)^\beta \leq c_\beta(a^\beta + b^\beta)$ with $c_\beta = \max(1, 2^{\beta-1})$ (subadditivity of $t^\beta$ for $\beta \leq 1$, convexity for $\beta > 1$). Hence $\|x - y\|^\beta \leq c_\beta(\|x\|^\beta + \|y\|^\beta)$, so for $P, Q \in \mathcal{P}_\beta$ all expectations in (1) and below are finite.

**Definition 3** (Energy distance [2, Def. 1, Prop. 2])**.** For $P, Q \in \mathcal{P}_\beta$ and independent $X, X' \sim P$, $Y, Y' \sim Q$,

$$D_\beta(P, Q) = 2\mathbb{E}\|X - Y\|^\beta - \mathbb{E}\|X - X'\|^\beta - \mathbb{E}\|Y - Y'\|^\beta. \tag{2}$$

**Lemma 4** (Fourier representation of $\|x\|^\beta$; [2, Lemma 1, eq. (5.2)])**.** *Let $0 < \beta < 2$. There is a constant $C(d, \beta) \in (0, \infty)$ such that for all $x \in \mathbb{R}^d$*

$$\int_{\mathbb{R}^d} \frac{1 - \cos\langle t, x\rangle}{\|t\|^{d+\beta}}\, dt = C(d, \beta)\, \|x\|^\beta, \qquad C(d, \beta) = \frac{2\pi^{d/2}\, \Gamma(1 - \beta/2)}{\beta\, 2^\beta\, \Gamma\big(\frac{d+\beta}{2}\big)}. \tag{3}$$

*Proof. Convergence.* The integrand is $\geq 0$. Using $1 - \cos u \leq \min(u^2/2, 2)$ and $|\langle t, e\rangle| \leq \|t\|$ for a unit vector $e$: near $t = 0$ the integrand is at most $\frac{1}{2}\|t\|^{2-d-\beta}$, and in polar coordinates $\int_0^1 r^{2-d-\beta} r^{d-1} dr = \int_0^1 r^{1-\beta} dr < \infty$ because $\beta < 2$; for $\|t\| \geq 1$ it is at most $2\|t\|^{-d-\beta}$ and $\int_1^\infty r^{-d-\beta} r^{d-1} dr = \int_1^\infty r^{-1-\beta} dr < \infty$ because $\beta > 0$. So the integral $I(x)$ is finite for every $x$. It is also strictly positive for $x \neq 0$, since the integrand is continuous and positive on a non-empty open set (e.g. near $t = \pi x/\|x\|^2$).

*Homogeneity.* For $x \neq 0$, substitute $t = s/\|x\|$, $dt = \|x\|^{-d} ds$:

$$I(x) = \|x\|^\beta \int \frac{1 - \cos\langle s, x/\|x\|\rangle}{\|s\|^{d+\beta}}\, ds = \|x\|^\beta I(x/\|x\|).$$

*Rotation invariance.* For an orthogonal matrix $R$, substituting $t = R^\top s$ (Jacobian 1, $\|t\| = \|s\|$) gives $I(Rx) = I(x)$. Every unit vector is a rotation of $e_1$, so $I(x/\|x\|) = I(e_1) =: C(d, \beta) \in (0, \infty)$, and $I(0) = 0$. This proves (3) with some finite positive constant, which is all the proof of Theorem 5 needs. The closed form of $C(d, \beta)$ is from [2, eq. (5.2), Appendix A]. We did not re-derive it; we checked it by quadrature for $d \in \{1, 2, 3\}$ and $\beta \in \{0.5, 1, 1.5\}$ (Section 9, Table 2). It also matches the constant in the characteristic-function form of the energy score [1, eq. (24)], which is $1/(2C(d, \beta))$ (they write $m$ for the dimension). □

**Theorem 5** (Energy distance characterises distributions; [2, Props. 1–2])**.** *Let $0 < \beta < 2$, $P, Q \in \mathcal{P}_\beta$, with characteristic functions $\varphi_P, \varphi_Q$. Then*

$$D_\beta(P, Q) = \frac{1}{C(d, \beta)} \int_{\mathbb{R}^d} \frac{|\varphi_P(t) - \varphi_Q(t)|^2}{\|t\|^{d+\beta}}\, dt. \tag{4}$$

*Consequently $D_\beta(P, Q) \geq 0$, with equality if and only if $P = Q$.*

*Proof.* Let $X, X' \sim P$, $Y, Y' \sim Q$ be independent. Apply Lemma 4 to $x = X - Y$, $X - X'$, $Y - Y'$ and take expectations. The integrands are non-negative, so Tonelli's theorem lets us swap $\mathbb{E}$ and $\int dt$; each of the three resulting quantities is finite by the moment condition. Since

$\mathbb{E}\cos\langle t, X-Y\rangle = \operatorname{Re}\mathbb{E}e^{i\langle t,X\rangle}\overline{\mathbb{E}e^{i\langle t,Y\rangle}} = \operatorname{Re}(\varphi_P(t)\overline{\varphi_Q(t)})$ and similarly $\mathbb{E}\cos\langle t, X-X'\rangle = |\varphi_P(t)|^2$, $\mathbb{E}\cos\langle t, Y-Y'\rangle = |\varphi_Q(t)|^2$, we get

$$C(d,\beta)\, D_\beta(P,Q) = \int \frac{2\big(1-\operatorname{Re}\varphi_P\overline{\varphi_Q}\big) - \big(1-|\varphi_P|^2\big) - \big(1-|\varphi_Q|^2\big)}{\|t\|^{d+\beta}}\, dt = \int \frac{|\varphi_P-\varphi_Q|^2}{\|t\|^{d+\beta}}\, dt,$$

using $|a-b|^2 = |a|^2 + |b|^2 - 2\operatorname{Re}(a\bar{b})$. The right side is $\geq 0$. If it is 0, then $\varphi_P = \varphi_Q$ Lebesgue-a.e., because the weight $\|t\|^{-d-\beta}$ is positive. Characteristic functions are continuous, so $\varphi_P = \varphi_Q$ everywhere, and $P = Q$ by the uniqueness theorem for characteristic functions. □

**Corollary 6** (Strict propriety)**.** *For $\beta \in (0,2)$, $S_\beta$ is strictly proper relative to $\mathcal{P}_\beta$.*

*Proof.* Immediate from Proposition 8 below and Theorem 5. □

**Proposition 7** ($\beta = 2$ is proper but not strictly proper)**.** *Let $P, Q \in \mathcal{P}_2$ have means $m_P, m_Q$ and covariances $\Sigma_P, \Sigma_Q$. Then*

$$S_2(P,y) = \|m_P - y\|^2, \qquad S_2(P,Q) = \|m_P - m_Q\|^2 + \operatorname{tr}\Sigma_Q, \qquad D_2(P,Q) = 2\|m_P - m_Q\|^2.$$

*So $S_2$ is proper, and any $P$ with $m_P = m_Q$ attains the minimum $S_2(Q,Q)$; it is not strictly proper.* Counterexample: *$P = \mathcal{N}(0,1)$, $Q = \mathcal{N}(0,4)$ in $d = 1$ have $D_2(P,Q) = 0$ but $P \neq Q$, while $D_1(P,Q) = (2\sqrt{10}-6)/\sqrt{\pi} \approx 0.1831 > 0$. Even the point mass $P = \delta_{m_Q}$ is optimal for $S_2$.*

*Proof.* For independent $X \sim P$, $Y \sim Q$: $\mathbb{E}\|X-y\|^2 = \operatorname{tr}\Sigma_P + \|m_P - y\|^2$, $\mathbb{E}\|X-X'\|^2 = 2\operatorname{tr}\Sigma_P$, $\mathbb{E}\|X-Y\|^2 = \operatorname{tr}\Sigma_P + \operatorname{tr}\Sigma_Q + \|m_P - m_Q\|^2$. Substitute into (1) and (2). For the Gaussian pair, $X-Y \sim \mathcal{N}(0,5)$, $X-X' \sim \mathcal{N}(0,2)$, $Y-Y' \sim \mathcal{N}(0,8)$ and $\mathbb{E}|\mathcal{N}(0,s^2)| = s\sqrt{2/\pi}$, so $D_1 = 2\sqrt{10/\pi} - 2/\sqrt{\pi} - 4/\sqrt{\pi} = (2\sqrt{10}-6)/\sqrt{\pi}$. □

In the Fourier picture, $C(d,\beta) \to \infty$ as $\beta \to 2$ (the factor $\Gamma(1-\beta/2)$ has a pole), so (4) degenerates; this matches [2, Prop. 2(ii)], $\mathcal{E}^{(2)}(X,Y) = 2|\mathbb{E}X - \mathbb{E}Y|^2$.

## 4 Expected score minus its minimum is half the energy distance

**Proposition 8.** *For $\beta \in (0,2)$ and $P, Q \in \mathcal{P}_\beta$,*

$$S_\beta(P,Q) - S_\beta(Q,Q) = \tfrac{1}{2}D_\beta(P,Q), \qquad S_\beta(Q,Q) = \tfrac{1}{2}\mathbb{E}\|Y-Y'\|^\beta.$$

*Proof.* $S_\beta(P,Q) = \mathbb{E}\|X-Y\|^\beta - \frac{1}{2}\mathbb{E}\|X-X'\|^\beta$ (the inner expectation in (1) is over $X, X' \sim P$, independent of $Y \sim Q$). With $P = Q$: $S_\beta(Q,Q) = \mathbb{E}\|Y-Y'\|^\beta - \frac{1}{2}\mathbb{E}\|Y-Y'\|^\beta = \frac{1}{2}\mathbb{E}\|Y-Y'\|^\beta$. Subtract and compare with (2). □

Hence, for a parametric model, $S_\beta(P_\theta, Q) = \frac{1}{2}D_\beta(P_\theta, Q) + \frac{1}{2}\mathbb{E}\|Y-Y'\|^\beta$. The second term does not depend on $\theta$, so minimising the expected energy score over $\theta$ is *the same optimisation problem* as minimising the energy distance to the data. The minimum value of $D_\beta$ is 0 and is attained exactly when $P_\theta = Q$ (Theorem 5).

## 5 The loss for the pushforward model

**Assumption 9.** $Q \in \mathcal{P}_\beta$ and $P_\theta \in \mathcal{P}_\beta$, i.e. $\mathbb{E}_{Z\sim\mu}\|F_\theta(Z)\|^\beta < \infty$. For images in $[-1,1]^d$, $Q$ has bounded support. If $\|F_\theta(z)\| \leq a_\theta + b_\theta\|z\|$ (true for MLPs and convolutional nets with Lipschitz activations), all Gaussian moments are finite. For DiT the output is a linear head applied to a modulated LayerNorm, so for each fixed class it is bounded. (This is an assumption about the architecture, not a proved property of a specific network.)

**Unconditional loss.** By the change-of-variables formula of Section 1,

$$\begin{aligned}\mathcal{L}_\beta(\theta) = S_\beta(P_\theta, Q) &= \mathbb{E}_{Z\sim\mu,\, Y\sim Q}\|F_\theta(Z) - Y\|^\beta - \tfrac{1}{2}\,\mathbb{E}_{Z,Z'\overset{\text{iid}}{\sim}\mu}\|F_\theta(Z) - F_\theta(Z')\|^\beta \\ &= \tfrac{1}{2}D_\beta(P_\theta, Q) + \tfrac{1}{2}\mathbb{E}\|Y - Y'\|^\beta,\end{aligned} \tag{5}$$

with $Z$, $Z'$, $Y$ mutually independent. $F_\theta(Z)$ and $F_\theta(Z')$ are then two independent draws from $P_\theta$, as (1) requires.

**Class-conditional loss.** The conditional model is the family $\{P_\theta^c\}_c$; it does not model $C$, whose law $\pi$ is given by the data. Define the conditional score $S_\beta^{\text{cond}}(\{P^c\}, (y, c)) = S_\beta(P^c, y)$ and

$$\mathcal{L}_\beta^{\text{cond}}(\theta) = \mathbb{E}_{(Y,C)\sim Q}\,\mathbb{E}_{Z,Z'\overset{\text{iid}}{\sim}\mu}\Big[\|F_\theta(Z,C) - Y\|^\beta - \tfrac{1}{2}\|F_\theta(Z,C) - F_\theta(Z',C)\|^\beta\Big] = \sum_c \pi_c\, S_\beta(P_\theta^c, Q_c). \tag{6}$$

Here $(Z, Z')$ is independent of $(Y, C)$, and *both generated samples use the same label* $C$.

**Proposition 10** (The conditional version is strictly proper)**.** *Let $\beta \in (0, 2)$ and $P^c, Q_c \in \mathcal{P}_\beta$ for all $c$. Then*

$$\mathbb{E}_{(Y,C)\sim Q} S_\beta^{\text{cond}}(\{P^c\}, (Y, C)) - \mathbb{E}_{(Y,C)\sim Q} S_\beta^{\text{cond}}(\{Q_c\}, (Y, C)) = \tfrac{1}{2}\sum_c \pi_c\, D_\beta(P^c, Q_c) \;\geq 0,$$

*with equality if and only if $P^c = Q_c$ for every $c$ with $\pi_c > 0$. Equivalently, viewed as a score for the joint law $\pi \otimes P^{\cdot}$ on $\mathbb{R}^d \times \{1..K\}$ with the label marginal fixed to $\pi$, it is strictly proper.*

*Proof.* Condition on $C = c$ (tower property) and apply Proposition 8 to each class. A finite sum of non-negative terms with weights $\pi_c$ is zero iff every term with $\pi_c > 0$ is zero; apply Theorem 5 to each. Two joint laws with the same label marginal $\pi$ coincide iff their conditionals agree for $\pi$-a.e. $c$. □

*Remark* 11 (Class weights). Replacing $\pi_c$ by any weights $w_c > 0$, e.g. class-balanced batches, does not move the minimiser in the well-specified case: each class term is minimised separately at $P^c = Q_c$. The weights only change the trade-off between classes when the model cannot fit all of them.

**Proposition 12** (Pooling the interaction term across classes is wrong)**.** *Suppose the second term pairs generated samples drawn with* independent *labels $C, C' \sim \pi$, i.e. $\mathcal{L}^{\text{pool}} = \sum_c \pi_c \mathbb{E}\|X_c - Y_c\|^\beta - \frac{1}{2}\mathbb{E}_{C,C'}\mathbb{E}\|X_C - X'_{C'}\|^\beta$ with $X_c \sim P^c$. Then*

$$\mathcal{L}^{\text{pool}} = \sum_c \pi_c\Big[\mathbb{E}\|X_c - Y_c\|^\beta - \tfrac{\pi_c}{2}\mathbb{E}\|X_c - X'_c\|^\beta\Big] - \tfrac{1}{2}\sum_{c\neq c'} \pi_c\pi_{c'}\mathbb{E}\|X_c - X'_{c'}\|^\beta.$$

*Within a class, the repulsion weight drops from 1 to $\pi_c < 1$. An extra cross-class repulsion appears, and it does not involve the data. This is not $\sum_c \pi_c S_\beta(P^c, Q_c)$, and in general $\{Q_c\}$ does not minimise it.*

*Proof.* The identity follows by expanding $\mathbb{E}_{C,C'}$ over the $K^2$ label pairs. To show that $\{Q_c\}$ is not a minimiser, take $d = 1$, $\beta = 1$, $K = 2$, $\pi_1 = \pi_2 = \frac{1}{2}$, $Q_1 = \mathcal{N}(-a, 1)$, $Q_2 = \mathcal{N}(a, 1)$ with $a > 0$, and the model family $P^1 = \mathcal{N}(-a, s^2)$, $P^2 = \mathcal{N}(a, s^2)$; $s = 1$ is the truth. Using $\mathbb{E}|\mathcal{N}(\delta, \sigma^2)| = \sigma\sqrt{2/\pi}\, e^{-\delta^2/2\sigma^2} + \delta(1 - 2\Phi(-\delta/\sigma))$, whose derivative in $\sigma$ is $\sqrt{2/\pi}\, e^{-\delta^2/2\sigma^2}$:

$$\mathcal{L}^{\text{pool}}(s) = \sqrt{2/\pi}\sqrt{1 + s^2} - \frac{s}{2\sqrt{\pi}} - \frac{1}{4}\,\mathbb{E}|\mathcal{N}(2a, 2s^2)|, \qquad \frac{d\mathcal{L}^{\text{pool}}}{ds}\Big|_{s=1} = \frac{1 - e^{-a^2}}{2\sqrt{\pi}} > 0.$$

So shrinking every class below its true spread lowers $\mathcal{L}^{\text{pool}}$, and the truth is not even a stationary point. (For $a = 0$ the classes coincide and pooling is harmless.) Check (d) (Section 9) shows the same under-dispersion in a trained model. □

## 6 Line-by-line check of the proposed formula

As typed in `CLAUDE.md`:

$$\text{Loss}(F_\theta) = \mathbb{E}_y\big[\mathbb{E}_x\|F_\theta(x) - y\|\big] - \tfrac{1}{2}\cdot\mathbb{E}_{x,x'}\big[F_\theta(x) - F_\theta(x')\big], \tag{7}$$

"$y$ from the original dataset, $x$ starts from a Gaussian, and $x$ and $x'$ both from the natural distribution". We compare it with (5), item by item.

**(1) Nested versus joint expectation: correct.** If $x \sim \mu$ is drawn independently of $y \sim Q$, Tonelli's theorem (integrand $\geq 0$) gives $\mathbb{E}_y[\mathbb{E}_x\|F_\theta(x) - y\|] = \mathbb{E}_{(x,y)\sim\mu\otimes Q}\|F_\theta(x) - y\|$. The nested form is the same quantity and is exactly the first term of (5). In the conditional case the nesting is $\mathbb{E}_{(y,c)}\mathbb{E}_x\|F_\theta(x,c) - y\|$, with $x$ independent of $(y,c)$. What matters is independence: if $x$ were coupled to $y$ (for example $x$ an encoding or a noised copy of $y$), the term would be a reconstruction error and no longer the energy score of $P_\theta$.

**(2) The second term is a raw difference: wrong.** $F_\theta(x) - F_\theta(x') \in \mathbb{R}^d$ is a vector, so as typed (7) subtracts a vector from a scalar. Read coordinate-wise, for $x, x'$ i.i.d. and $\mathbb{E}\|F_\theta(x)\| < \infty$,

$$\mathbb{E}_{x,x'}[F_\theta(x) - F_\theta(x')] = \mathbb{E}F_\theta(x) - \mathbb{E}F_\theta(x') = 0 \in \mathbb{R}^d.$$

Any scalar reduction (sum or mean over coordinates, or the norm of this expectation) is 0 too. In code, `(F(x)-F(x2)).mean()` has expectation 0, and its gradient has mean 0: it only adds noise. So, read literally, (7) is

$$\text{Loss}_{\text{literal}}(\theta) = \mathbb{E}\|F_\theta(x) - y\| = \mathbb{E}_{X\sim P_\theta} g(X), \qquad g(a) := \mathbb{E}_{Y\sim Q}\|a - Y\|.$$

**Proposition 13** (The literal reading collapses)**.** $\inf_{P\in\mathcal{P}_1}\mathbb{E}_{X\sim P}g(X) = \min_a g(a)$. *The minimisers are exactly the distributions supported on* $\arg\min g$*, the set of geometric (spatial) medians of* $Q$*. If* $Q$ *is not concentrated on a line,* $g$ *is strictly convex, the median* $a^*$ *is unique, and the only minimiser is the point mass* $\delta_{a^*}$*: every* $z$ *is mapped to the same image.*

*Proof.* $g$ is convex, finite and continuous, with $g(a) \geq \|a\| - \mathbb{E}\|Y\| \to \infty$, so $\arg\min g \neq \emptyset$. $\mathbb{E}_P g(X) \geq \min g$, with equality iff $P(g(X) = \min g) = 1$. For strict convexity, take $a \neq b$ and the line $\ell$ through them. For $y \notin \ell$, the vectors $a - y$ and $b - a$ are linearly independent, so equality never holds in the triangle inequality along $t \mapsto (a - y) + t(b - a)$, and $t \mapsto \|a + t(b - a) - y\|$ is strictly convex. If $Q(\ell) < 1$, integrating over $y$ gives strict convexity of $g$ on $[a, b]$. ($g$ is 1-Lipschitz, hence continuous.) □

Engression [4, Sec. 2.3] makes the same remark: minimising the first term alone targets the (conditional) median, a deterministic model. **Fix:** the second term must be the expected *norm* of the difference, $\frac{1}{2}\mathbb{E}\|F_\theta(x) - F_\theta(x')\|$, raised to the same power $\beta$ as the first term.

**(3) Which distribution $x, x'$ come from: must be the source Gaussian, independent.** In (1), $X, X'$ are two independent draws from the *model* $P_\theta$. For a pushforward model this means $X = F_\theta(x)$ and $X' = F_\theta(x')$ with $x, x' \overset{\text{iid}}{\sim} \mathcal{N}(0, I)$, independent of $y$ (and, conditionally, sharing the label of $y$). Other readings of "the natural distribution" fail:

- $x, x'$ drawn from the data, with no $F_\theta$: the term is $\frac{1}{2}\mathbb{E}\|y - y'\|$, a constant. Its gradient is 0, so we are back to Proposition 13.
- $x, x'$ drawn from the data, but passed through $F_\theta$: the term $\mathbb{E}\|F_\theta(y) - F_\theta(y')\|$ evaluates the network on inputs it never sees at sampling time. It is not a function of $P_\theta$, so the loss is not a scoring rule for the model. We reject this reading.
- $x' = x$ (re-using the same noise draw): $\|F_\theta(x) - F_\theta(x)\| = 0$, again collapse.
- Conditional model with the labels of $x$ and $x'$ drawn independently: wrong minimiser (Proposition 12).

**(4) "Euclidean distance": norm ($\beta = 1$), not squared norm ($\beta = 2$).** The first term of (7) uses $\|\cdot\|$, i.e. $\beta = 1$, which is strictly proper (Corollary 6). With squared norms in both terms ($\beta = 2$), Proposition 7 gives $\mathcal{L}_2(\theta) = \|m_{P_\theta} - m_Q\|^2 + \operatorname{tr}\Sigma_Q$: only the mean is fitted, and the shape, spread and modes of $P_\theta$ get no training signal. With the squared norm in the first term alone, $\mathbb{E}\|X - Y\|^2 = \|m_P - m_Q\|^2 + \operatorname{tr}\Sigma_P + \operatorname{tr}\Sigma_Q$ is minimised by $\delta_{m_Q}$, the mean image: the familiar "blurry average". Mixed powers are not proper either. For example, take $\|\cdot\|$ in the first term and $\|\cdot\|^2$ in the second, with $X = sU$ for a non-degenerate $U$:

$$\mathbb{E}\|sU - Y\| - \tfrac{s^2}{2}\mathbb{E}\|U - U'\|^2 \le s\,\mathbb{E}\|U\| + \mathbb{E}\|Y\| - \tfrac{s^2}{2}\mathbb{E}\|U - U'\|^2 \xrightarrow[s\to\infty]{} -\infty,$$

so that loss is unbounded below and training diverges. Both terms must use the same power $\beta \in (0, 2)$.

**(5) The factor $\frac{1}{2}$ and the sign: both correct.** Consider the weighted family $\mathcal{L}^\lambda(P) = \mathbb{E}\|X - Y\|^\beta - \frac{\lambda}{2}\mathbb{E}\|X - X'\|^\beta$.

**Proposition 14.** *Let $\beta \in (0, 2)$ and $Q \in \mathcal{P}_\beta$ not a point mass.*

*(i)* $\lambda = 1$ *(the $\frac{1}{2}$ in (7)): strictly proper; the unique minimiser is $Q$.*

*(ii)* $0 \le \lambda < 1$*: not strictly proper in general. For $Q = \mathcal{N}(m, \sigma^2 I_d)$, the best Gaussian $\mathcal{N}(m', s^2 I_d)$ has $m' = m$ and $s^2 = \sigma^2/(2\lambda^{-2/(2-\beta)} - 1) < \sigma^2$ [5, Prop. 4.1]. For $\beta = 1$ this is $s^2 = \sigma^2\lambda^2/(2 - \lambda^2)$ (we re-derived it for $d = 1$, and check (b3) confirms it numerically). The model is under-dispersed.*

*(iii)* $\lambda > 1$*, which includes coefficient 1 instead of $\frac{1}{2}$ ($\lambda = 2$): $\inf_{P\in\mathcal{P}_\beta} \mathcal{L}^\lambda(P) = -\infty$. There is no minimiser, and training diverges.*

*(iv) Plus sign instead of minus ($\lambda < 0$): both terms are minimised by a point mass. Collapse.*

*Proof.* (i) is Corollary 6. (ii): in $d = 1$, $\beta = 1$, $Q = \mathcal{N}(0, 1)$, $P = \mathcal{N}(0, s^2)$: $\mathcal{L}^\lambda = \sqrt{2/\pi}\sqrt{1 + s^2} - \lambda s/\sqrt{\pi}$. Setting the derivative to zero gives $\sqrt{2}\,s/\sqrt{1 + s^2} = \lambda$, i.e. $s^2 = \lambda^2/(2 - \lambda^2)$. The general $(d, \beta)$ formula is from [5]. (iii): let $P_s = (1 - p)\delta_0 + p\,\delta_{sv}$ with $\|v\| = 1$, $p \in (0, 1)$. Then $\mathbb{E}\|X - X'\|^\beta = 2p(1 - p)s^\beta$, and by dominated convergence (dominating function $c_\beta(1 + \|Y\|^\beta)$)

$\mathbb{E}\|sv - Y\|^\beta = s^\beta(1 + o(1))$ as $s \to \infty$. So $\mathcal{L}^\lambda(P_s) = (1-p)\mathbb{E}\|Y\|^\beta + p\, s^\beta\big[1 + o(1) - \lambda(1-p)\big]$. If $\lambda > 1$, pick $p < 1 - 1/\lambda$; then the bracket tends to $1 - \lambda(1-p) < 0$ and $\mathcal{L}^\lambda(P_s) \to -\infty$. (iv): $-\frac{\lambda}{2}\mathbb{E}\|X - X'\|^\beta \geq 0$, with equality for point masses, and the first term $\mathbb{E}_P g_\beta(X)$, $g_\beta(a) = \mathbb{E}\|a - Y\|^\beta$, is minimised by a point mass at a minimiser of the continuous, coercive $g_\beta$ (the argument of Proposition 13). □

Within the Gaussian scale family the loss stays bounded until $\lambda \geq \sqrt{2}$ (for $\beta = 1$); see check (b3). The unboundedness in (iii) uses non-Gaussian $P_s$, which a flexible network can approximate.

**(6) Summary: what each reading minimises.**

| reading of (7) | minimiser over all $P$ |
|---|---|
| literal: $-\frac{1}{2}\mathbb{E}[F(x) - F(x')]$ (raw difference $= 0$) | point mass at the geometric median of $Q$ (Prop. 13) |
| $x, x'$ from the data: $-\frac{1}{2}\mathbb{E}\|y - y'\|$ (constant) | same as literal |
| same noise draw, $x' = x$ | same as literal |
| squared norms in both terms ($\beta = 2$) | any $P$ with the mean of $Q$, e.g. $\delta_{m_Q}$ (Prop. 7) |
| squared norm in the first term only | $\delta_{m_Q}$ (the mean image) |
| $\|\cdot\|$ then $-\frac{1}{2}\|\cdot\|^2$ | none: unbounded below |
| coefficient 1 instead of $\frac{1}{2}$ | none: unbounded below (Prop. 14(iii)) |
| coefficient $\frac{\lambda}{2}$, $\lambda < 1$ | under-dispersed (Prop. 14(ii)) |
| labels of $x, x'$ drawn independently | under-dispersed within each class (Prop. 12) |
| **norm, $\frac{1}{2}$, $x, x' \overset{\text{iid}}{\sim} \mathcal{N}(0, I)$ indep. of $y$, same label** | $\boldsymbol{Q}$ **(unique; Cor. 6, Prop. 10)** |

**The loss (the specification for the code).**

$$\boxed{\mathcal{L}(\theta) = \mathbb{E}_{(Y,C)\sim Q}\, \mathbb{E}_{Z,Z' \overset{\text{iid}}{\sim}\mathcal{N}(0,I)}\Big[\|F_\theta(Z,C) - Y\|^\beta - \tfrac{1}{2}\|F_\theta(Z,C) - F_\theta(Z',C)\|^\beta\Big], \quad \beta = 1,} \tag{8}$$

with $(Z, Z')$ independent of $(Y, C)$. Then $\mathcal{L}(\theta) = \frac{1}{2}\sum_c \pi_c D_\beta(P_\theta^c, Q_c) + \text{const}$. The unconditional loss drops $C$. Any $\beta \in (0, 2)$ is strictly proper; $\beta = 1$ is the default (Section 7).

**Verdict.** The proposed formula is *not correct as written*, but it is one symbol away from the energy score. The first term, the minus sign and the factor $\frac{1}{2}$ are right. The second term must be the expected *norm* $\frac{1}{2}\mathbb{E}\|F_\theta(x) - F_\theta(x')\|$, with $x, x'$ i.i.d. from the *source Gaussian* (not the data), independent of $y$, and with the same label as $y$ in the conditional case. As typed, the second term is identically zero, and the minimiser collapses to a single point.

## 7 Minibatch estimator

**Batch structure.** A batch has $N$ labelled data points split into groups $g = 1, \ldots, G$. All points in group $g$ share the label $c_g$. The group holds $n_g \geq 1$ data points $y_{g,1..n_g}$ and $m_g \geq 2$ generated samples $x_{g,j} = F_\theta(z_{g,j}, c_g)$, with $z_{g,j} \overset{\text{iid}}{\sim} \mathcal{N}(0, I)$ drawn fresh and independent of the data. The estimator is

$$\widehat{\mathcal{L}} = \sum_{g=1}^{G} \frac{n_g}{N}\left[\frac{1}{n_g m_g}\sum_{i=1}^{n_g}\sum_{j=1}^{m_g}\|x_{g,j} - y_{g,i}\|^\beta - \frac{1}{2\, m_g(m_g - 1)}\sum_{j\neq k}\|x_{g,j} - x_{g,k}\|^\beta\right]. \tag{9}$$

Two special cases matter. *Per-class groups*: group the batch by label, so $G \leq K$, and pair every generated sample of class $c$ with every data point of class $c$. *Per-data-point groups*: $n_g = 1$, $m$ samples per data point, which is the engression estimator [4, eq. (13)]. The unconditional case is one group, $G = 1$.

**Proposition 15** (Unbiasedness)**.** *If the $N$ labelled data points are i.i.d. from $Q$ and grouped by label (any grouping that depends only on the labels), then $\mathbb{E}\widehat{\mathcal{L}} = \mathcal{L}_\beta^{\mathrm{cond}}(\theta)$. With class-stratified batches, $\mathbb{E}\widehat{\mathcal{L}} = \sum_c w_c S_\beta(P_\theta^c, Q_c)$ for the sampling weights $w_c$, which has the same minimiser (Remark 11).*

*Proof.* Condition on the labels and group sizes. Within group $g$, each $(x_{g,j}, y_{g,i})$ is an independent pair from $P_\theta^{c_g} \otimes Q_{c_g}$, and each $(x_{g,j}, x_{g,k})$ with $j \neq k$ is an independent pair from $P_\theta^{c_g}$. So the bracket has conditional mean $S_\beta(P_\theta^{c_g}, Q_{c_g})$; this needs $m_g \geq 2$, since the diagonal $j = k$ is excluded. The conditional mean of $\widehat{\mathcal{L}}$ is therefore $\sum_c (N_c/N) S_\beta(P_\theta^c, Q_c)$, where $N_c$ is the number of class-$c$ points. Since $\mathbb{E}N_c/N = \pi_c$, the claim follows. □

Keeping the diagonal (a V-statistic, dividing by $m_g^2$) biases the interaction term down by a factor $(m_g - 1)/m_g$. That is the same as using $\lambda = (m_g - 1)/m_g < 1$ in Proposition 14(ii), i.e. an under-dispersed minimiser; for $m_g = 2$ this is $\lambda = \frac{1}{2}$.

**What to log.** (9) minus an estimate of $\frac{1}{2}\mathbb{E}\|Y - Y'\|^\beta$ (per class) estimates $\frac{1}{2}D_\beta$. The full U-statistic

$$\widehat{D}_\beta = \frac{2}{nm}\sum_{i,j}\|x_j - y_i\|^\beta - \frac{1}{m(m-1)}\sum_{j\neq k}\|x_j - x_k\|^\beta - \frac{1}{n(n-1)}\sum_{i\neq i'}\|y_i - y_{i'}\|^\beta$$

is unbiased for $D_\beta(P, Q)$. It can be negative for a single batch, but its expectation is $\geq 0$ and is 0 iff $P = Q$ (check (a1)). Use it, per class, as the held-out metric.

**Variance as $m$ changes.** Take one group with $n$ data points and $m$ generated samples. Projecting (9) onto single samples (the first step of the Hoeffding decomposition of a U-statistic) gives

$$\widehat{\mathcal{L}} - \mathbb{E}\widehat{\mathcal{L}} \approx \frac{1}{n}\sum_i \big(a(y_i) - \mathbb{E}a\big) + \frac{1}{m}\sum_j \big(\phi(x_j) - \psi(x_j) - \mathbb{E}[\phi - \psi]\big),$$

with $a(y) = \mathbb{E}\|X - y\|^\beta$, $\phi(x) = \mathbb{E}\|x - Y\|^\beta$ and $\psi(x) = \mathbb{E}\|x - X'\|^\beta$; the interaction term contributes $2 \times \frac{1}{2}\psi$. So $\operatorname{Var}\widehat{\mathcal{L}} = \sigma_Y^2/n + \tau^2/m + O(1/(nm) + 1/m^2)$, with $\tau^2 = \operatorname{Var}_{X\sim P}[\phi(X) - \psi(X)]$. Two consequences. (i) For fixed $n$, increasing $m$ only shrinks the second term, so the variance plateaus at $\sigma_Y^2/n$. (ii) At the optimum $P = Q$, we have $\phi \equiv \psi$ and $\tau^2 = 0$: near convergence the generated-sample noise is of second order. The same first-order argument applies to the gradient. Check (a4) (Figure 1) shows the plateau, and shows that at equal compute, pooling all generated samples of a group against all of its data points (one group, $m = M/n$) has lower variance than per-data-point groups. This is the main reason to prefer per-class grouping in stage 4.

**Unbiased gradients.** If $\theta \mapsto F_\theta(z)$ is locally Lipschitz with a constant integrable in $z$, then $\theta \mapsto \|F_\theta(z) - y\|$ is too. $P_\theta$ is atomless whenever $F_\theta(Z)$ has a density, and then the non-differentiable events $F_\theta(z) = y$ and $F_\theta(z) = F_\theta(z')$ have probability 0. Dominated convergence then gives $\nabla_\theta \mathbb{E}\widehat{\mathcal{L}} = \mathbb{E}\nabla_\theta \widehat{\mathcal{L}}$: SGD on (9) follows unbiased gradients. These conditions are an assumption about the network, not checked for DiT. Bellemare et al. [7] make the contrast with the Wasserstein metric, whose sample gradients are biased.

**The gradient of $\|\cdot\|$ at 0 ($\beta = 1$).** $\|u\|$ is not differentiable at $u = 0$. Its subdifferential there is the closed unit ball, and 0 is the natural (minimum-norm) subgradient. Zero distances occur in three places:

1. The diagonal $j = k$ of the generated-pair matrix. It must be excluded anyway (unbiasedness). Computing $\sqrt{\sum(x_j - x_k)^2}$ for the full matrix and *then* masking the diagonal gives `NaN` gradients, because $\partial\sqrt{s}/\partial s = \infty$ at $s = 0$ and $\infty \cdot 0 =$ `NaN` (check (e)).

2. DiT's adaLN-Zero initialisation zeroes the final linear layer (`initialize_weights` in the reference code's `models.py`, commit `ed81ce2` [10]). At step 0, $F_\theta(z, c) = 0$ for every $z$, so *every* generated pair is at distance exactly 0. A naive implementation produces `NaN` at the first step. With the safe norm, the interaction gradient at step 0 is 0; the confinement gradient alone moves the zero-initialised head, and features that depend on $z$ break the symmetry (check (f)).

3. Exact duplicates $x_j = x_k$ after collapse. These have probability 0 while $P_\theta$ is atomless.

*How to handle it:* compute $\|u\|^\beta$ as `where(s>0, where(s>0,s,1)**(beta/2), 0)` with $s = \|u\|^2$ (the "double-where"). This gives the exact value and gradient 0 at $u = 0$, with no smoothing bias. `torch.linalg.vector_norm` and `torch.cdist` also return gradient 0 at 0 in PyTorch 2.14.1 (check (e)). But `torch.cdist`'s default matrix-multiplication path computes the distance of identical 3072-dimensional rows as about 0.04 rather than 0 in float32, so exclude the diagonal by index, never by relying on it being 0. For $\beta \in (1, 2)$, $\|u\|^\beta$ is differentiable at 0 with gradient 0. For $\beta < 1$ the gradient is unbounded near 0, which is one more reason to use $\beta = 1$. A smoothed norm $\sqrt{\|u\|^2 + \varepsilon^2}$ is not needed.

**Compute cost.** Per step, with $M = \sum_g m_g$ generated samples: $M$ network forward and backward passes, which dominate. The distances cost $O\big(\sum_g (n_g m_g + m_g^2)\, d\big)$ operations. For $N = M = 256$ and $d = 3072$ that is about $4 \times 10^8$ multiply-adds, negligible next to a DiT pass. Memory is two $M \times N$ and $M \times M$ scalar matrices. Use $\|a\|^2 + \|b\|^2 - 2\langle a, b\rangle$ (or `cdist`) rather than broadcasting an $M \times N \times d$ tensor, which would need 0.8 GB for these sizes. Per-class grouping with $m_c = n_c$ (one generated sample per data point) costs the same network passes as ordinary supervised training, and needs at least two data points per class present in the batch. Alternatively, draw $r \geq 2$ samples per data point.

**Recommendation for stage 4.** $\beta = 1$. Group by class within the batch, with weights $n_c/N$. Use $m_c \geq 2$ generated samples per class (simplest: one or two per data point, or class-stratified batches). Exclude the diagonal by index, and use the double-where norm. Log the per-class held-out $\widehat{D}_1$. Engression used $m = 2$ generated samples per observation per step [4, App. E].

## 8 Related work

**Energy distance is an MMD.** Sejdinovic, Sriperumbudur, Gretton and Fukumizu [3] show that for a semimetric $\rho$ of negative type, the distance-induced kernel $k(z, z') = \frac{1}{2}[\rho(z, z_0) + \rho(z', z_0) - \rho(z, z')]$ [3, Def. 13] gives $D_{E,\rho}(P, Q) = 2\gamma_k^2(P, Q)$, twice the squared MMD [3, Thm. 22]. With $\rho(x, y) = \|x - y\|^\beta$, our loss is (up to constants) an MMD loss with the distance kernel. One-step generators trained with MMD go back at least to generative moment matching networks [8], an MLP generator trained with MMD in one forward pass. Gneiting and Raftery's kernel scores [1, Sec. 5.1, Thm. 4, Ex. 9] put the energy score in the same family.

**Engression.** Shen and Meinshausen [4] fit a conditional generator $g(x, \varepsilon)$ by minimising $\mathbb{E}\big[\|Y - g(X,\varepsilon)\| - \frac{1}{2}\|g(X,\varepsilon) - g(X,\varepsilon')\|\big]$ [4, eq. (12)], with the unbiased estimator normalised by $2m(m-1)$ [4, eq. (13)]. This is exactly (8) with $\beta = 1$, covariates $X$ in place of the class $C$, and an MLP in place of DiT. Our test is engression for class-conditional image generation, with a DiT backbone. Gritsenko et al. [9] trained a parallel speech synthesiser with a generalised energy distance on spectrograms, without adversarial training.

**Scoring-rule (distributional) diffusion.** De Bortoli et al. [5] replace the regression loss of diffusion models with a conditional generalised energy score: a network $\hat{x}_\theta(t, x_t, \xi)$, $\xi \sim \mathcal{N}(0, I)$, models $p_{0|t}(\cdot \mid x_t)$ and is trained with $S_{\lambda,\beta}(p^\theta_{0|t}(\cdot|x_t), x_0) = \frac{\lambda}{2}\mathbb{E}\|X - X'\|^\beta - \mathbb{E}\|X - x_0\|^\beta$ [5, eq. (12)], integrated over $t$ with a weight $w_t$ [5, eq. (13)]. The score is strictly proper for $\lambda = 1$, $\beta \in (0, 2)$; $\beta \to 2$, $\lambda \to 0$ recovers the usual regression loss. Their reported gains are largest when sampling with few steps. In image models the noise $\xi$ is concatenated with $x_t$ along the channel dimension. Shen, Meinshausen and Zhang [6] chain several engression models along a forward noising process (reverse Markov learning).

**How our test relates, and the next step.** Our one-step map is the endpoint of that family: a single noise level, and $x_t$ is pure noise carrying no information about $x_0$. So the model must learn the whole marginal $Q$ (per class) from $z$ alone, which is the hardest case for a one-step model. The natural next step is the conditional energy score at several noise levels: the input becomes $(x_t, t, \xi)$, with $x_t$ the forward-noised data and $\xi$ fresh noise. The loss is (8) with the conditioning $(c)$ replaced by $(c, x_t, t)$, weighted over $t$, as in [5]. Strict propriety for each $(x_t, t)$ follows from Proposition 10 with the label replaced by $(x_t, t)$; for a continuous conditioning variable the argument is the same, with sums replaced by integrals. DiT already has the timestep input for this.

## 9 Numerical checks

All code is in `docs/stage1_checks/`: `run_all.sh` runs everything, and `es_core.py` holds the estimators. Results are in `results/*.json`. Environment: conda env `tmap-tex` (conda-forge: Python 3.11, NumPy 2.4.6, SciPy 1.17.1, CPU-only PyTorch 2.14.1), on 8 CPU cores. Total runtime 17 minutes.

### (a) The estimator

**(a1)** $d = 2$, $n = m = 32$, 4000 replicates. For $P = Q = \mathcal{N}(0, I)$, the U-statistic $\widehat{D}_1$ has mean $0.0014 \pm 1.0 \times 10^{-3}$ (one s.e.) and is negative in 61% of replicates. The V-statistic (diagonal kept) has mean 0.112, biased upward. For $P = \mathcal{N}((0.5, 0), I)$, $\widehat{D}_1$ has mean $0.1102 \pm 0.0020$, against the closed form 0.1099. That value comes from $2\mathbb{E}\|X - Y\| - \mathbb{E}\|X - X'\| - \mathbb{E}\|Y - Y'\|$, with $\|X - Y\|$ Rice-distributed (`scipy.stats.rice`; cross-checked by a $4 \times 10^6$-sample Monte Carlo estimate, 0.1095).

**(a2)** Closed form: for $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$, $X - Y \sim \mathcal{N}(\delta, s^2)$ with $\delta = \mu_1 - \mu_2$, $s^2 = \sigma_1^2 + \sigma_2^2$, and $\mathbb{E}|X - Y| = s\sqrt{2/\pi}\, e^{-\delta^2/2s^2} + \delta\,(1 - 2\Phi(-\delta/s))$ (folded-normal mean; checked against quadrature to $2 \times 10^{-15}$). So $D_1 = 2\mathbb{E}|X - Y| - 2\sigma_1/\sqrt{\pi} - 2\sigma_2/\sqrt{\pi}$. Table 1 compares it with the estimator mean ($n = m = 64$, 3000 replicates).

**(a3)** Table 2: quadrature of $\int (1 - \cos\langle t, e_1\rangle)\|t\|^{-d-\beta} dt$ (radial form, truncated at $r = 4000$ with an analytic tail) against $C(d, \beta)$ in (3). Also, for $P = \mathcal{N}(0, 1)$, $Q = \mathcal{N}(1, 4)$, the characteristic-function integral (4) gives 0.5341053241, against the closed form 0.5341053241.

Table 1: Check (a2): estimator mean vs closed-form $D_1$ for 1-D Gaussians.

| $P$ | $Q$ | closed-form $D_1$ | estimator mean $\pm$ s.e. | $z$ |
|---|---|---|---|---|
| $\mathcal{N}(0, 1^2)$ | $\mathcal{N}(0, 1^2)$ | 0.0000 | -0.0001 $\pm$ 0.0005 | -0.20 |
| $\mathcal{N}(0, 1^2)$ | $\mathcal{N}(0.5, 1^2)$ | 0.1396 | 0.1394 $\pm$ 0.0019 | -0.10 |
| $\mathcal{N}(0, 1^2)$ | $\mathcal{N}(0, 2^2)$ | 0.1831 | 0.1851 $\pm$ 0.0016 | +1.28 |
| $\mathcal{N}(1, 0.5^2)$ | $\mathcal{N}(-1, 1.5^2)$ | 2.0533 | 2.0560 $\pm$ 0.0061 | +0.44 |
| $\mathcal{N}(0, 1^2)$ | $\mathcal{N}(3, 1^2)$ | 3.7777 | 3.7901 $\pm$ 0.0066 | +1.87 |

Table 2: Check (a3): the constant $C(d, \beta)$ of Lemma 4.

| $d$ | $\beta$ | $C(d, \beta)$, eq. (3) | quadrature | rel. error |
|---|---|---|---|---|
| 1 | 0.5 | 5.01325655 | 5.01325655 | $6 \times 10^{-10}$ |
| 1 | 1.0 | 3.14159265 | 3.14159265 | $2 \times 10^{-11}$ |
| 1 | 1.5 | 3.34217103 | 3.34217103 | $6 \times 10^{-11}$ |
| 2 | 0.5 | 12.01316876 | 12.01316898 | $2 \times 10^{-8}$ |
| 2 | 1.0 | 6.28318531 | 6.28318531 | $6 \times 10^{-10}$ |
| 2 | 1.5 | 5.84224320 | 5.84224320 | $2 \times 10^{-10}$ |
| 3 | 0.5 | 20.99947993 | 20.99947994 | $6 \times 10^{-10}$ |
| 3 | 1.0 | 9.86960440 | 9.86960440 | $2 \times 10^{-11}$ |
| 3 | 1.5 | 8.39979197 | 8.39979197 | $3 \times 10^{-10}$ |

**(a4)** Variance of the loss estimate and of its gradient, as a function of the number of generated samples, for $n = 64$ data points from the 8-Gaussian target and model $X = sZ$ at $s = 1$ (far from optimum, so $\tau^2 > 0$). Figure 1.

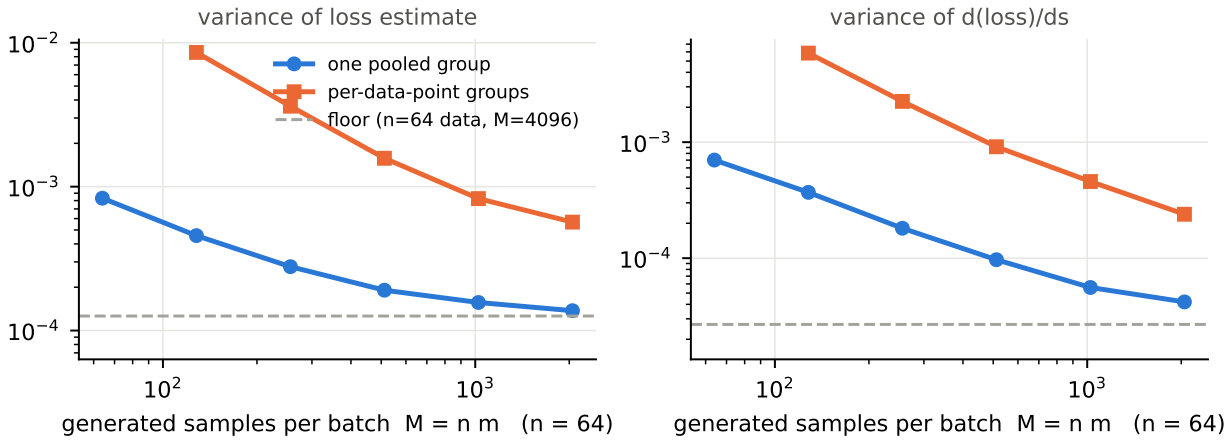



Figure 1: Check (a4). Variance vs. generated samples per batch $M = nm$ with $n = 64$ fixed. Both structures plateau at the data-only floor (dashed: $M = 4096$), as $\sigma_Y^2/n + \tau^2/m$ predicts. At equal $M$, one pooled group (blue) beats per-data-point groups (orange); for the loss estimate it is up to 19× lower.

### (b) $\beta = 2$ cannot separate equal means

$P = \mathcal{N}(0, 1)$, $Q = \mathcal{N}(0, 4)$, $n = m = 200$, 2000 replicates. $\widehat{D}_2$ has mean $0.002 \pm 0.0016$ (closed form 0). $\widehat{D}_1$ has mean $0.1833 \pm 0.0010$ (closed form 0.1831) and is positive in 100% of replicates. A mean shift of 0.5 is visible to $\beta = 2$: $\widehat{D}_2$ mean 0.512 against $2 \cdot 0.5^2 = 0.5$. Figure 2, right: expected score minus minimum over $P = \mathcal{N}(0, \sigma^2)$, in closed form using $\mathbb{E}|\mathcal{N}(0, s^2)|^\beta = s^\beta 2^{\beta/2}\Gamma(\frac{\beta+1}{2})/\sqrt{\pi}$. The minimum is at $\sigma = 2$ for every $\beta < 2$ and becomes flatter as $\beta \to 2$; at $\beta = 2$ the curve is identically 0. **(b3)** For $\mathbb{E}|X - Y| - \frac{\lambda}{2}\mathbb{E}|X - X'|$ over $X \sim \mathcal{N}(0, s^2)$, $Q = \mathcal{N}(0, 1)$, the numerical minimiser matches $s^2 = \lambda^2/(2 - \lambda^2)$ to $4 \times 10^{-8}$ for $\lambda \in \{0.25, \ldots, 1.25\}$. For $\lambda = 2$ the loss is $-330$ at $s = 1000$, decreasing without bound.

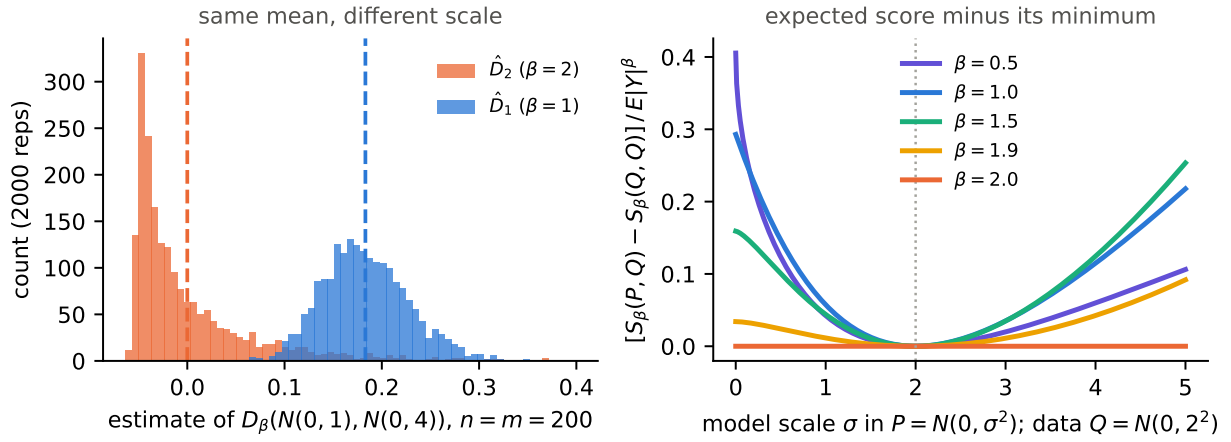



Figure 2: Check (b). Left: $\widehat{D}_2$ is centred at 0 although $P \neq Q$; $\widehat{D}_1$ is centred at the closed form (dashed). Right: normalised excess expected score; flat for $\beta = 2$.

### (c) Training a 2-D map with the corrected loss and with readings of the proposed one

MLP $2 \to 256 \to 256 \to 256 \to 2$ (SiLU), $z \sim \mathcal{N}(0, I_2)$, target: 8 Gaussians (radius 2, std 0.1). Adam, learning rate $10^{-3}$ with cosine decay to 0 (identical for every loss), 3000 steps, $n = M = 256$ per step, 3 seeds. Metric: $\widehat{D}_1$ between 4000 model samples and 4000 fresh target samples. Reference values: target vs. target $5.6 \times 10^{-4}$; source Gaussian vs. target 0.241; point mass at the origin (the geometric median) vs. target 1.47.

Table 3: Check (c): held-out $\widehat{D}_1$ to the target (mean $\pm$ sd over 3 seeds) and spread of the samples (seed 0).

| loss | held-out $\widehat{D}_1$ | sample std (x, y) | mean radius | outside $[-4, 4]^2$ |
|---|---|---|---|---|
| corrected loss, $\beta = 1$ | $0.00983 \pm 5.4 \times 10^{-4}$ | 1.43, 1.39 | 1.98 | 0% |
| corrected loss, $\beta = 1.5$ | $0.0159 \pm 0.0011$ | 1.42, 1.41 | 1.97 | 0% |
| literal: $-\frac{1}{2}$ `(F(x)-F(x')).mean()` | $1.46 \pm 0.0014$ | 0.00128, 0.00166 | 0.00285 | 0% |
| $x, x'$ from the data: $-\frac{1}{2}\mathbb{E}\|y - y'\|$ | $1.46 \pm 6.4 \times 10^{-4}$ | 0.000512, 0.00016 | 0.002 | 0% |
| squared norms in both terms ($\beta = 2$) | $1.46 \pm 0.0029$ | 0.0009, 0.00268 | 0.00247 | 0% |
| coefficient 1 instead of $\frac{1}{2}$ | $2.41 \times 10^8 \pm 7.0 \times 10^7$ | 3.32e+08, 3.38e+08 | 4.09e+08 | 100% |
| $\|\cdot\|$, then $-\frac{1}{2}\|\cdot\|^2$ | $9.65 \times 10^8 \pm 1.2 \times 10^7$ | 1.06e+09, 1.06e+09 | 1.34e+09 | 100% |
| target (8 Gaussians) | $5.6 \times 10^{-4}$ | 1.39, 1.44 | 2.00 | 0% |

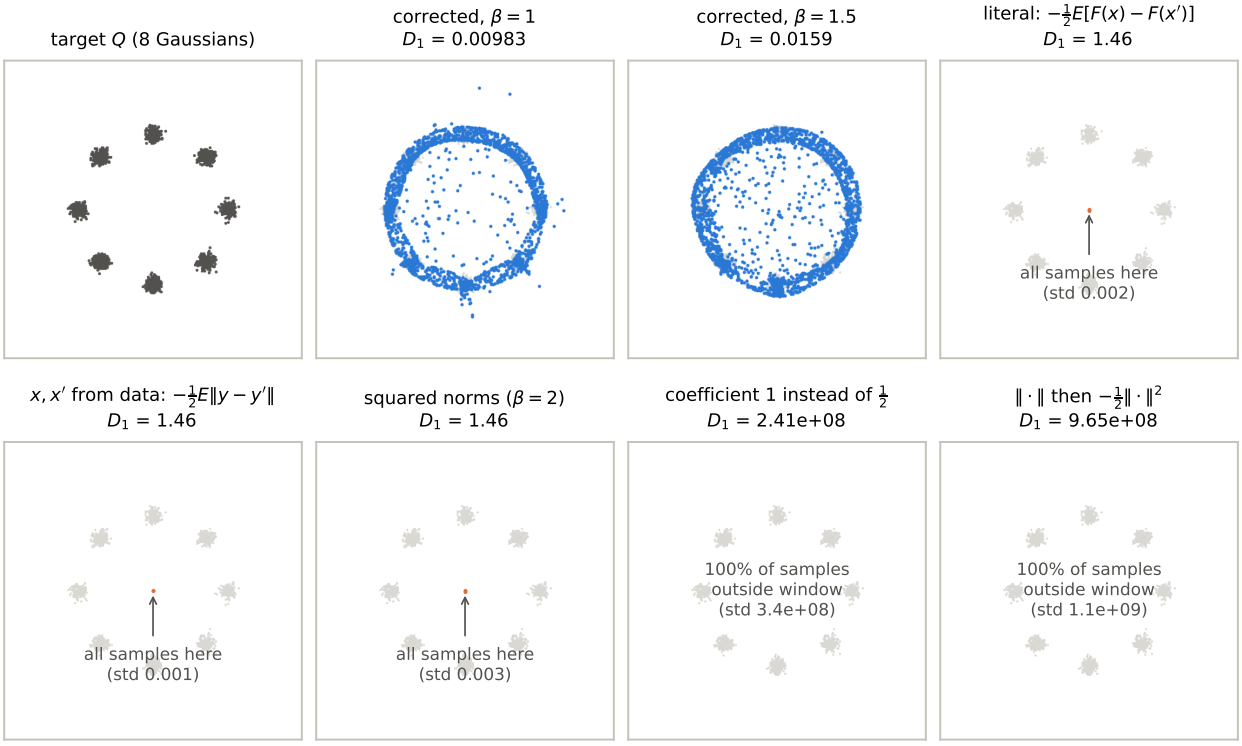



Figure 3: Check (c). Samples (seed 0, coloured) over the target (grey). Blue: strictly proper losses. Orange: readings of the proposed formula. Window $[-4, 4]^2$.

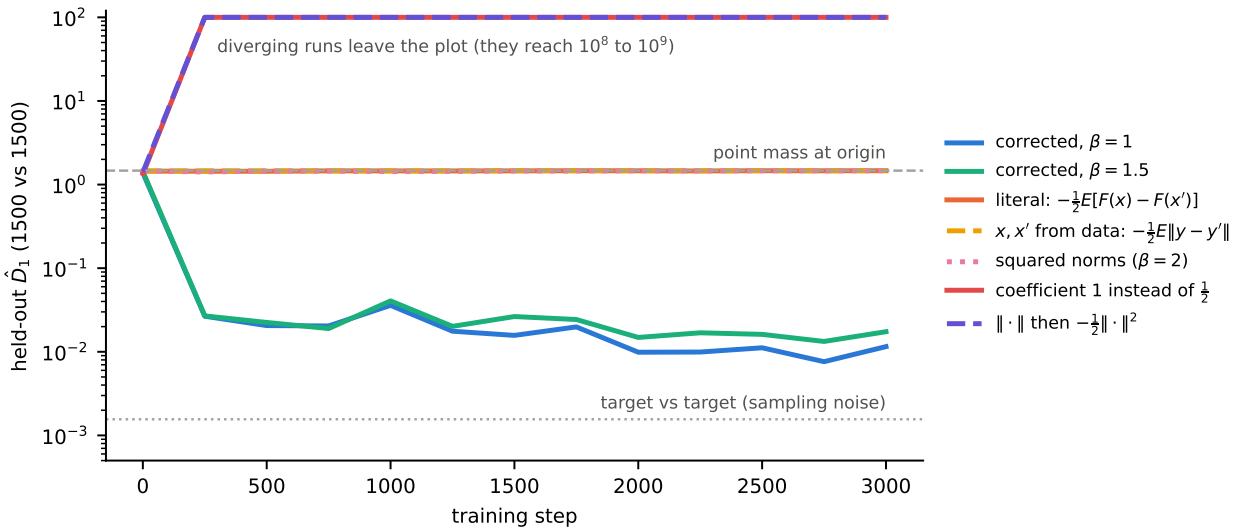



Figure 4: Check (c). $\widehat{D}_1$ to the target during training (seed 0, 1500 vs. 1500 samples; values are clipped to $[10^{-3}, 10^2]$, so negative estimates sit at the bottom and diverging runs at the top).

**Reading of check (c).** The two strictly proper losses learn the ring and its scale: $\widehat{D}_1 = 0.00983$ ($\beta = 1$) and 0.0159 ($\beta = 1.5$), against 0.241 for the untrained source Gaussian and 1.47 for a point mass. Their sample std, about 1.4 per coordinate, matches the target's 1.42. After 3000 steps, though, the eight modes are only partly separated: Figure 3 shows mass spread along the circle. Check (c2) (below) shows this is mainly a training-budget effect. Every reading of the proposed formula fails as Section 6 predicts:

- *Literal* (raw difference, implemented as `(F(x)-F(x2)).mean()`), $x, x'$ *from the data*, and *squared norms* ($\beta = 2$) all collapse to an almost-point mass near the centre. The sample std is between $3.7 \times 10^{-4}$ and 0.0025 for the literal reading, and $\widehat{D}_1 \approx 1.46$ equals the point-mass baseline. For $\beta = 2$ the expected loss is flat in every direction except the mean (Proposition 7); nothing holds the spread, and in all three seeds it drifted to a near point mass ($\widehat{D}_1 = 1.46$).
- *Coefficient* 1 *instead of* $\frac{1}{2}$ diverges: the minibatch loss reaches $-1.6 \times 10^8$, and the sample std grows to between $2.7 \times 10^8$ and $3.6 \times 10^8$ (Proposition 14(iii)). Growth slows only because the learning rate decays to 0.
- $\|\cdot\|$ *then* $-\frac{1}{2}\|\cdot\|^2$ is unbounded below (Section 6, item 4): minibatch loss down to $-2.6 \times 10^{18}$, $\widehat{D}_1 = 9.65 \times 10^8$.

**(c2) Longer training.** The corrected loss ($\beta = 1$, seed 0), trained the same way for 3000 and for 20000 steps. After 3000 steps, $\widehat{D}_1 = 0.00897$ and 49% of samples lie within 0.3 (three target standard deviations) of a mode centre. After 20000 steps, $\widehat{D}_1 = 6.0 \times 10^{-5}$ (below the target-vs-target sampling noise) and the fraction is 84%; the target has 99%. Figure 5. The loss does separate the modes, but the fine structure is learned much more slowly than the coarse structure. That is consistent with the weight $\|t\|^{-d-\beta}$ in (4), which puts most of $D_\beta$ on low frequencies. We did not test this explanation further. For images, expect fine detail to converge slowly (Section 10).

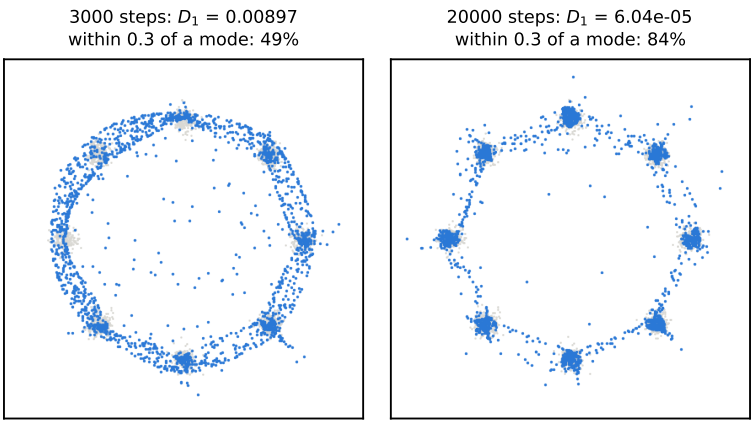



Figure 5: Check (c2). Corrected loss, $\beta = 1$, seed 0, after 3000 and 20000 steps.

### (d) Class-conditional grouping

This is the example from the proof of Proposition 12: $d = 1$, $Q_0 = \mathcal{N}(-3, 1)$, $Q_1 = \mathcal{N}(3, 1)$, $\pi = (\frac{1}{2}, \frac{1}{2})$, conditional MLP $F(z, \text{onehot}(c))$, $z \sim \mathcal{N}(0, 1)$, one generated sample per data point, 3000 steps, 2 seeds. Within the Gaussian family, the pooled loss has derivative 0.28206 at $s = 1$ (numerical), against $(1 - e^{-a^2})/(2\sqrt{\pi}) = 0.28206$. Its minimiser is $s^* = 0.378$, close to the $\lambda = \frac{1}{2}$ value $\sqrt{1/7} = 0.378$. Training gives: *grouped* (correct) per-class std 0.94–1.06 (target 1), $\widehat{D}_1 \leq 0.015$; *pooled* (bug) per-class std 0.60–0.65, $\widehat{D}_1$ between 0.39 and 0.63. The pooled model also pushes the class means apart, to |mean| between 3.72 and 3.95 instead of 3. That is the data-free cross-class repulsion in Proposition 12; the Gaussian-family calculation held the means fixed, which is why the trained spread differs from $s^*$. Either way the bug gives a wrong minimiser, and the trained model shows it. Figure 6.

*Secondary 2-D run* (`check_d2_conditional_2d.py`): two classes on the 8-Gaussian circle, interleaved (class 0 even modes, class 1 odd modes). $\widehat{D}_1$ per class, for seed 0 then seed 1: grouped $0.005, 0.006, 0.007, 0.006$; pooled $0.007, 0.003, 0.383, 0.336$. Per-coordinate RMS distance to the nearest mode centre (the target has 0.100; the bridges between modes inflate it for every model): grouped $0.209, 0.206, 0.204, 0.211$, pooled $0.142, 0.134, 0.168, 0.155$. The effect is smaller and seed-dependent here, because with interleaved classes the cross-class repulsion also depends on within-class spread. We report it for completeness. The 1-D case is the clean test of Proposition 12.

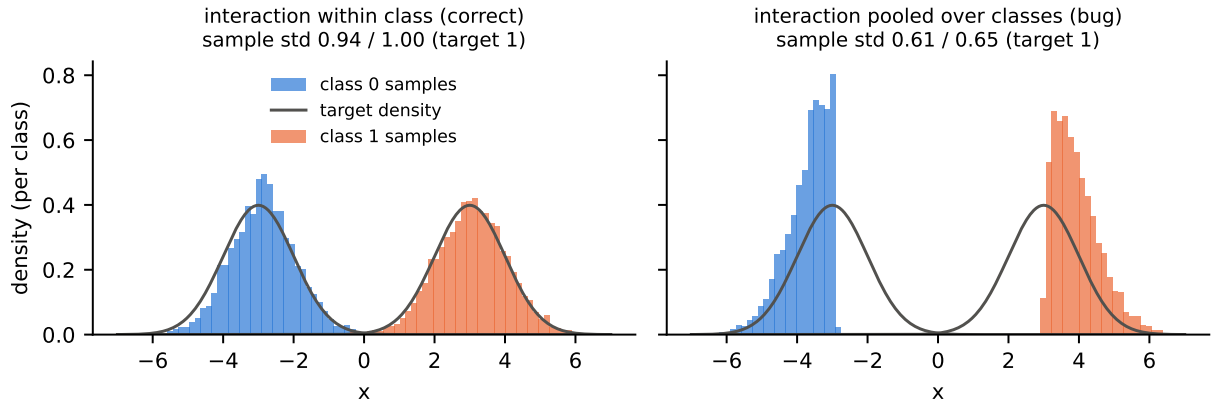



Figure 6: Check (d), seed 0. Per-class histograms of generated samples against the target densities (grey). Left: interaction term within each class (correct). Right: interaction pooled over all generated pairs regardless of class (Proposition 12): each class is pushed away from the other and under-dispersed.

### (e), (f) Gradient at zero

**(e)** PyTorch 2.14.1 on CPU, a $3 \times 2$ input whose rows 0 and 2 coincide, and the interaction term computed on the full $3 \times 3$ matrix with the diagonal masked by multiplication:

- `sqrt(((x[:,None]-x[None])**2).sum(-1))` then masking: `NaN` gradient for every input. The mask does not help, because $0 \cdot \texttt{NaN} = \texttt{NaN}$.
- `torch.linalg.vector_norm` and `torch.cdist`: finite gradients, 0 contribution from zero-distance pairs (`vector_norm` of an exact zero vector has gradient $(0, 0)$). `cdist(x,x)**0.5` was also finite here. Do not rely on this: it holds because `cdist`'s backward zeroes zero-distance entries.
- The double-where distance (`es_core.torch_pow_dist`): finite for $\beta = 0.5, 1, 1.5$.
- `torch.cdist` on $65 \times 3072$ float32 inputs (matrix-multiplication path) returns diagonal entries up to 0.04, and 0.03 for an exact duplicate pair. With the option

  `compute_mode="donot_use_mm_for_euclid_dist"`

  they are exactly 0.

**(f)** The 2-D toy MLP with a zero-initialised last layer, as in DiT, so every generated pair has distance exactly 0 at step 0. With the double-where distance, training is unaffected: 0 steps with `NaN` gradients, final $\widehat{D}_1 = 0.0126$. With the naive `sqrt`, the gradient is already `NaN` at the first step, 3000 of 3000 steps have `NaN` gradients, and the trained model outputs `NaN`: once a `NaN` reaches the Adam state, the parameters are lost.

## 10 Summary for the next stages

- **Loss:** (8) with $\beta = 1$: confinement $\|F_\theta(Z, C) - Y\|$ minus $\frac{1}{2}$ interaction $\|F_\theta(Z, C) - F_\theta(Z', C)\|$, with $Z, Z'$ i.i.d. $\mathcal{N}(0, I)$, independent of the data, and *the same class* in both interaction arguments.

- **Estimator:** (9), grouping by class within the batch, weights $n_c/N$, $m_c \geq 2$, diagonal excluded (U-statistic), double-where norm.
- **Metric:** per-class held-out $\widehat{D}_1$ (unbiased; can be $< 0$ on a batch).
- **Watch for:** `NaN` at step 0 with DiT's zero-initialised head if the norm is computed naively; V-statistic bias with small $m_c$; pooling the interaction across classes.

**Open questions.** (1) $\beta = 1$ versus $\beta \in (1, 2)$ on images: both are strictly proper; $\beta = 1$ is standard (engression, CRPS), and De Bortoli et al. report results over a range of $\beta$ [5]. Untested here. (2) Pixel space versus latent space: the theory is the same, but the Euclidean norm on pixels is a poor perceptual metric, and the energy distance in high dimension may be dominated by low-frequency or mean differences. This is the main practical risk for the one-step image test (stage 2 decides). Check (c2) points the same way: in 2-D the energy score learned coarse structure in a few hundred steps but needed about 20000 steps to separate the modes. (3) How small the per-class $m_c$ can be at DiT scale without the variance hurting; check (a4) suggests $m_c = n_c$ with per-class pooling is enough to reach the data-noise floor.

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